Introduction to 3D Vision

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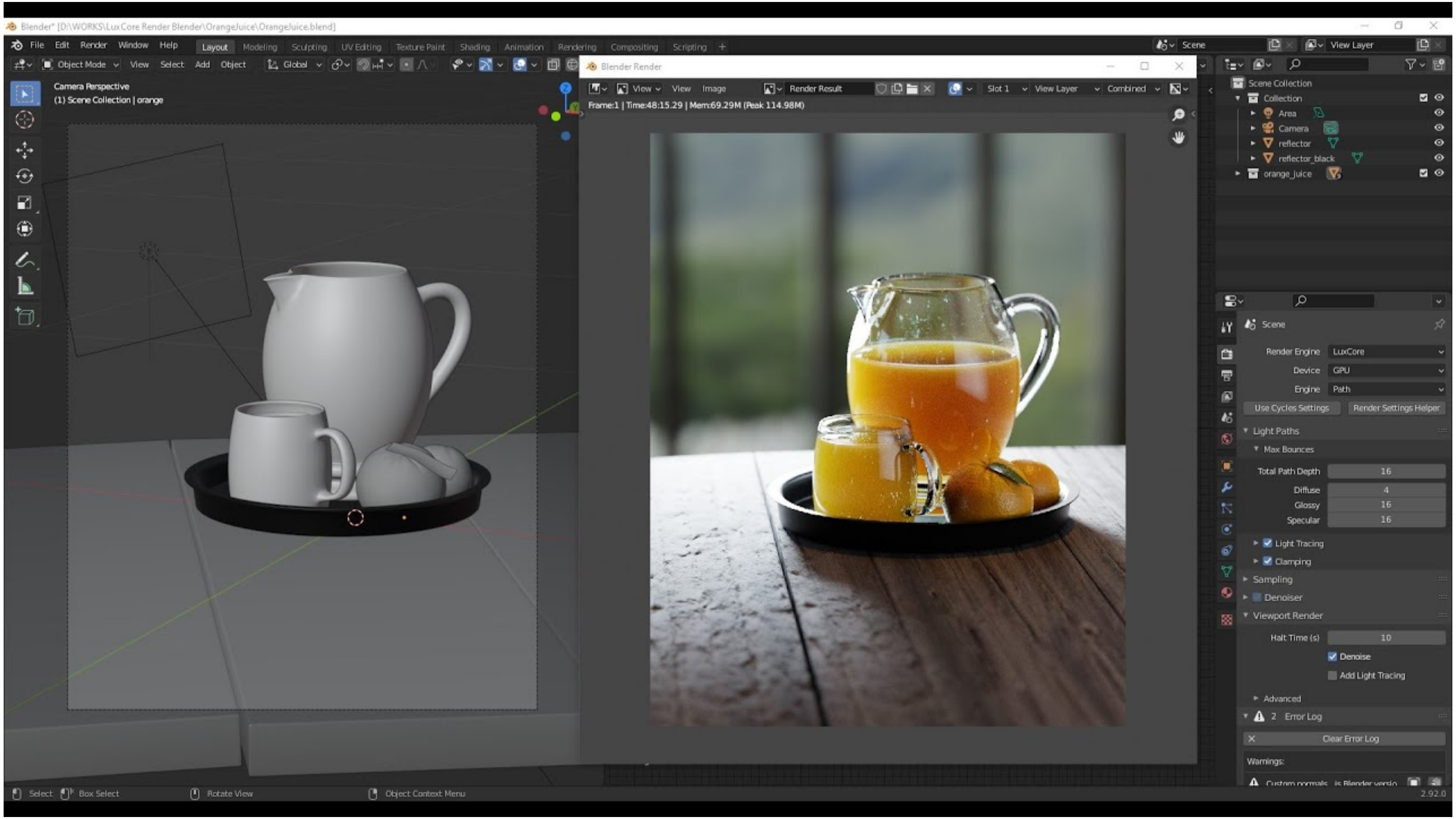
Why Two Eyes?

Rendering vs Vision



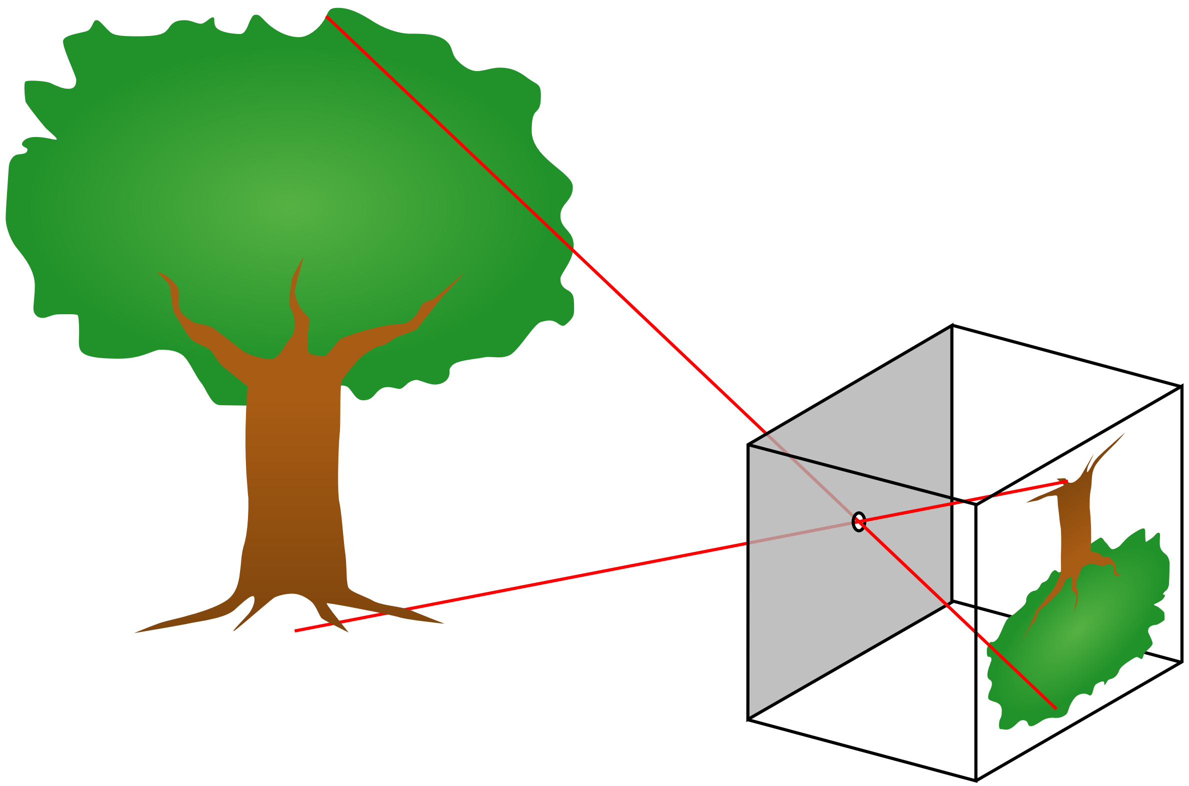
3D Vision Examples

Roadmap

1. Pinhole image formation
2. Homogeneous coordinates
3. Camera coordinates and pose
4. Intrinsics and the projection matrix
5. Calibration and distortion
6. Direct Linear Transform (DLT)

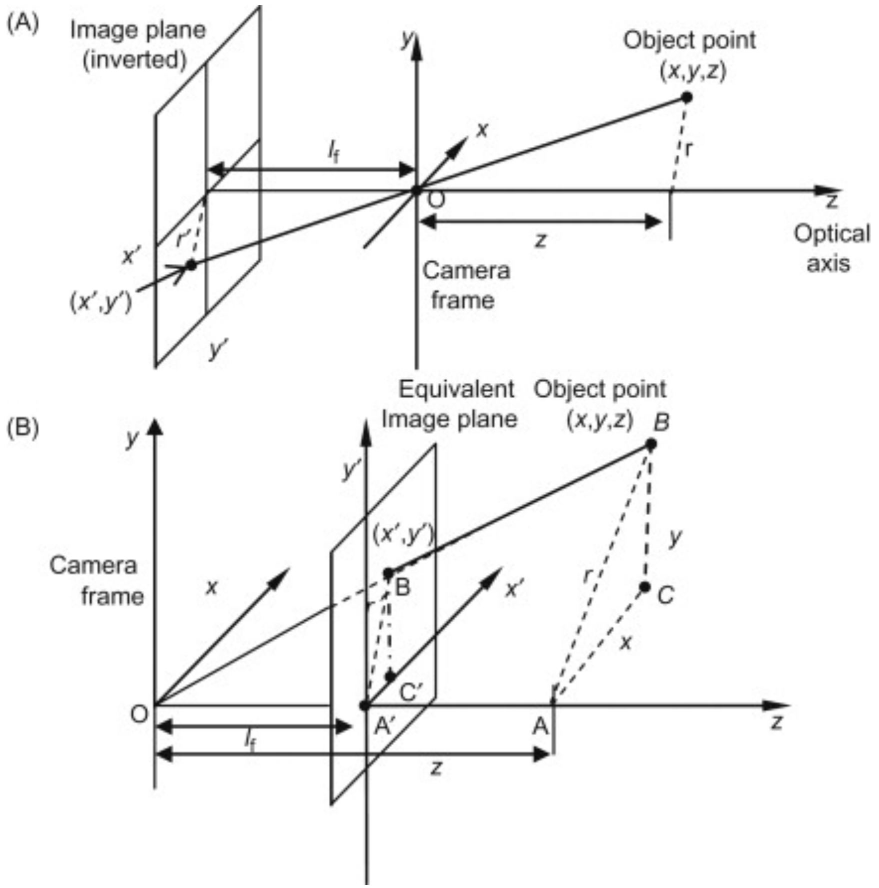
Image Formation

Pinhole Camera



- A pinhole camera performs **central projection**.
- Every scene point is projected along the ray through the camera center.
- The physical image plane behind the pinhole is inverted.

Similar Triangles



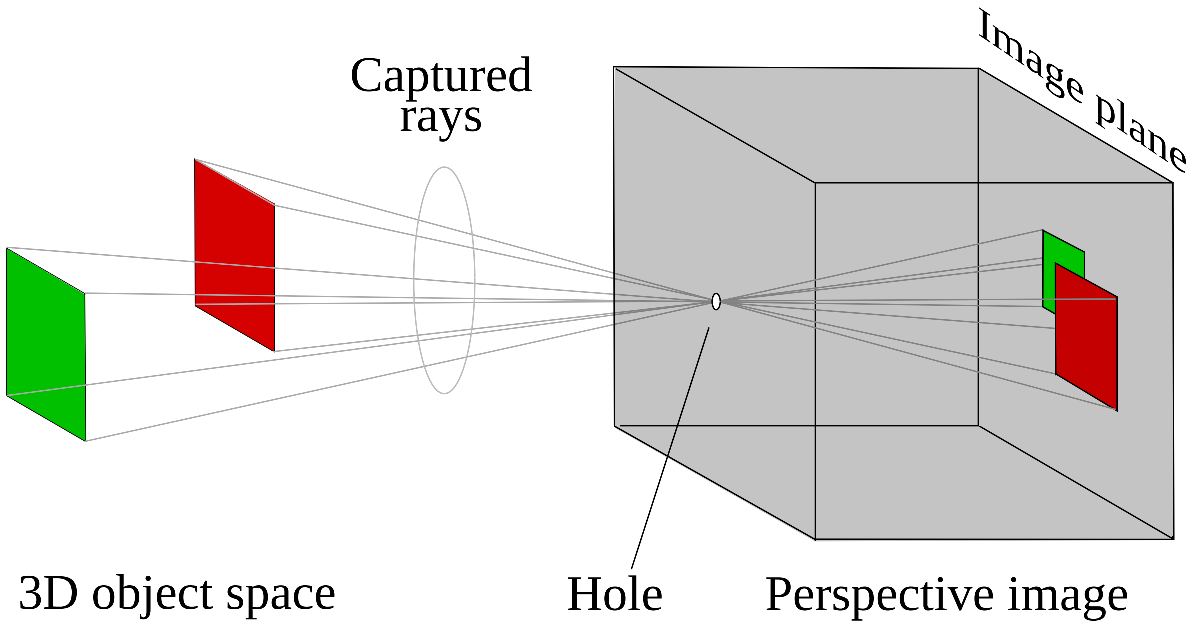
With the real image plane behind the center:

$$\frac{x'}{f} = -\frac{X_c}{Z_c}, \qquad \frac{y'}{f} = -\frac{Y_c}{Z_c}.$$

So

$$x' = -f\frac{X_c}{Z_c}, \qquad y' = -f\frac{Y_c}{Z_c}.$$

Virtual Image Plane

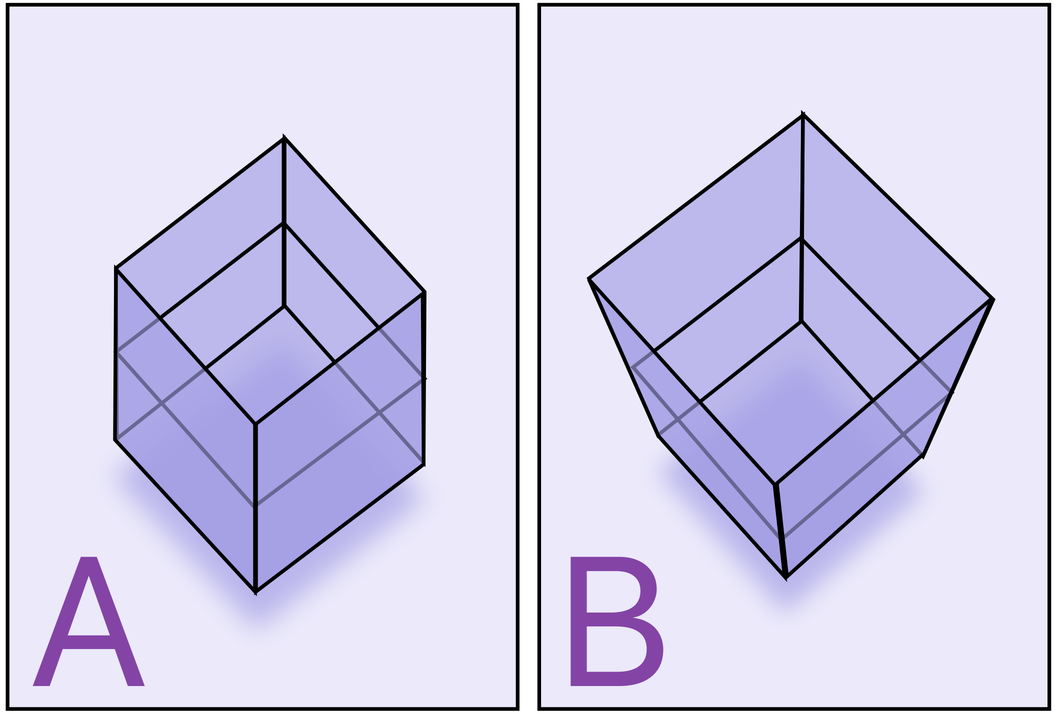


- Computer vision usually places the image plane in front of the camera.
- This removes the sign flip:

$$x = f\frac{X_c}{Z_c}, \qquad y = f\frac{Y_c}{Z_c}.$$

- The choice is a convention; consistency matters.

Perspective Foreshortening



- Projection divides by depth Z_c.
- Doubling the distance halves the projected size.
- The mapping is nonlinear in Euclidean coordinates:

$$(X_c, Y_c, Z_c) \mapsto \left(f\frac{X_c}{Z_c}, f\frac{Y_c}{Z_c} \right).$$

Homogeneous Coordinates

Euclidean image coordinates are obtained by dehomogenization:

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} \sim \begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{pmatrix}, \qquad u = \frac{\tilde{u}}{\tilde{w}}, \quad v = \frac{\tilde{v}}{\tilde{w}}.$$

The symbol $\sim$ means equality up to a nonzero scale.

Normalized Pinhole Projection

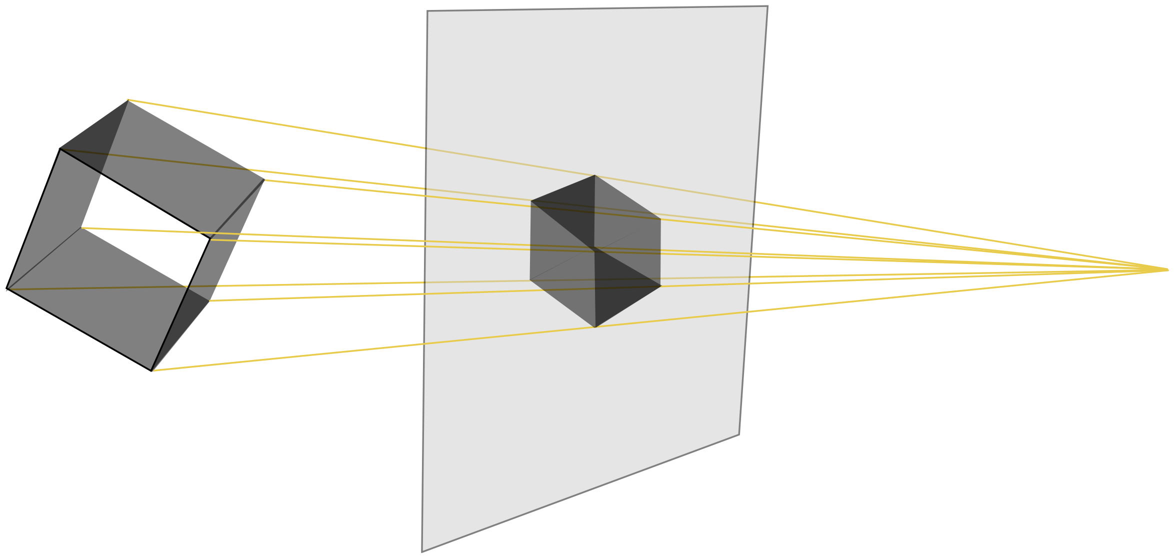
For a camera-frame point $\mathbf{X}_c = (X_c, Y_c, Z_c)^\top$:

$$\mathbf{x}_n = \begin{pmatrix} x_n \\ y_n \\ 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{pmatrix}.$$

After dehomogenization:

$$x_n = \frac{X_c}{Z_c}, \qquad y_n = \frac{Y_c}{Z_c}.$$

Perspective Projection



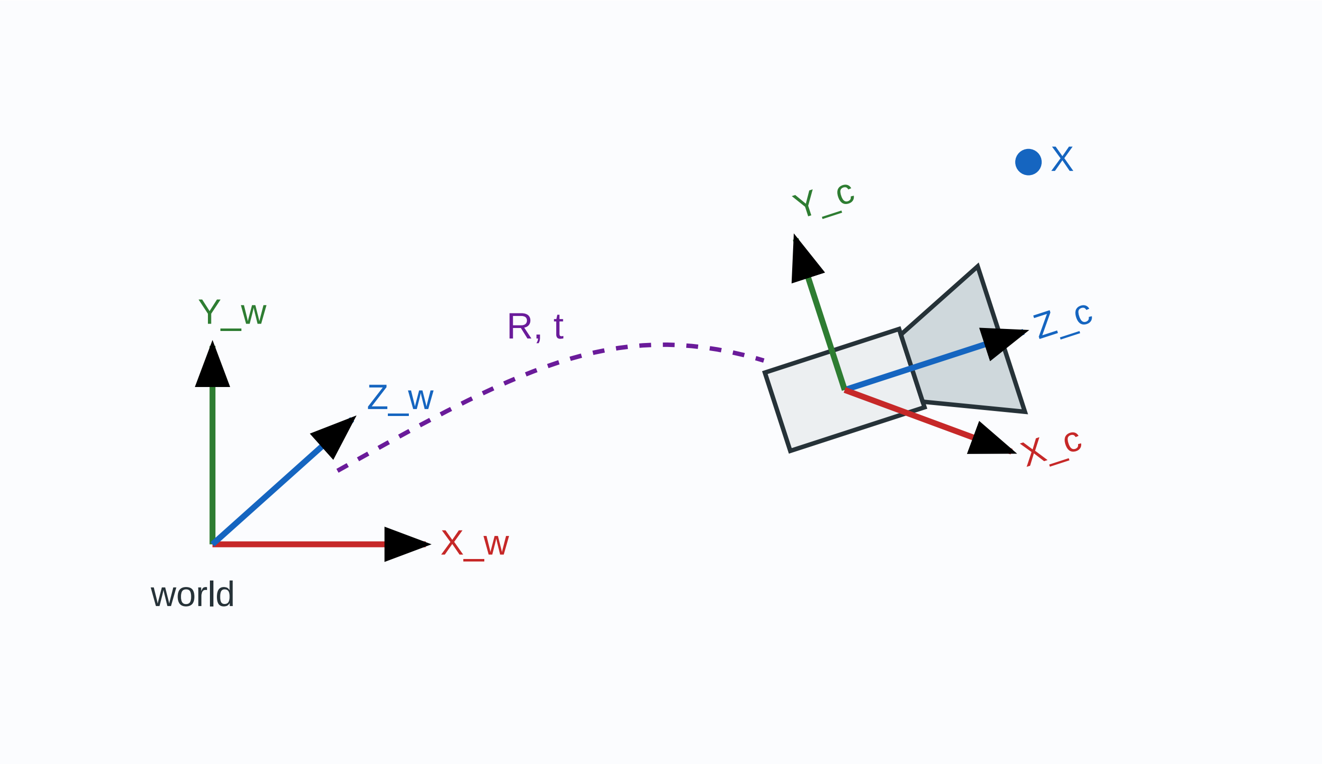
The homogeneous form is linear before dehomogenization:

$$\mathbf{x}_n \sim \Pi_0 \mathbf{X}_c, \Pi_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Depth is carried by the third homogeneous coordinate.

Camera Coordinates

World and Camera Frames



- World coordinates describe the scene.
- Camera coordinates place the origin at the optical center.
- The camera Z_c axis points along the optical axis.

Extrinsic Transformation

The world-to-camera transform is

$$\mathbf{X}_c = R\mathbf{X}_w + \mathbf{t}, \qquad R \in SO(3), \quad \mathbf{t} \in \mathbb{R}^3.$$

In homogeneous coordinates:

$$\begin{pmatrix} \mathbf{X}_c \\ 1 \end{pmatrix} = \begin{pmatrix} R & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{pmatrix} \begin{pmatrix} \mathbf{X}_w \\ 1 \end{pmatrix}.$$

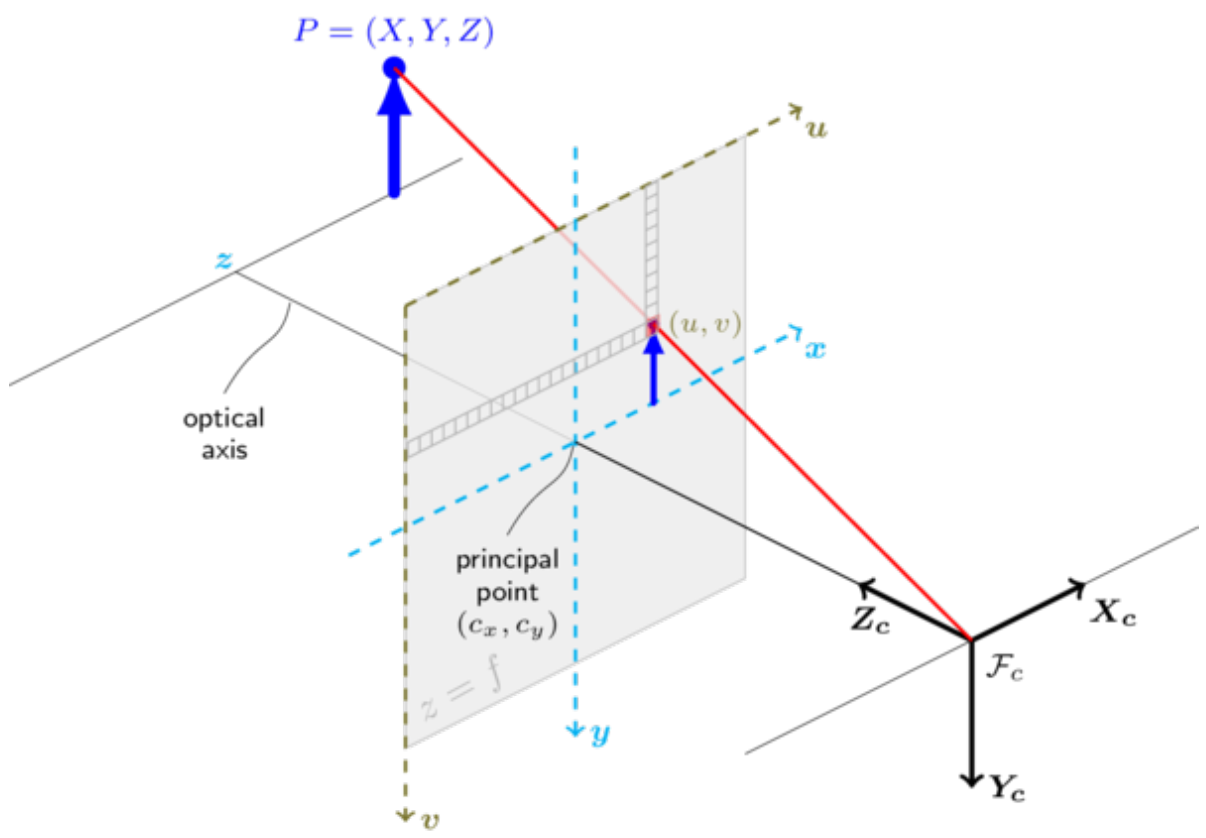
Intrinsics

Intrinsic Matrix

Pixel coordinates are obtained from normalized image coordinates:

$$\lambda \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = K \begin{pmatrix} x_n \\ y_n \\ 1 \end{pmatrix}.$$

$$K = \begin{pmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}.$$



Intrinsic Parameters

- f_x, f_y: focal length in horizontal and vertical pixel units.
- (c_x, c_y): principal point, where the optical axis meets the image plane.

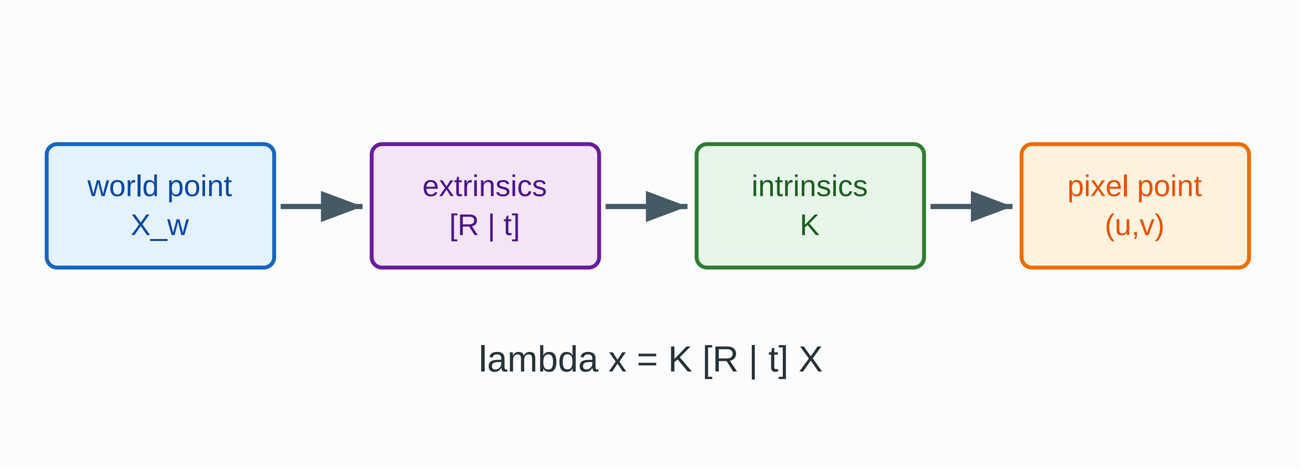
$$
\begin{aligned}
u &= f_x x_n + c_x\,, \\
v &= f_y y_n + c_y\,.
\end{aligned}
$$

For square pixels:

$$
K = \begin{pmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{pmatrix}.
$$

Projection Matrix

Full Camera Model



For a homogeneous world point $\mathbf{X} = (X, Y, Z, 1)^\top$:

$$\lambda \mathbf{x} = K[R \mid \mathbf{t}]\mathbf{X}, \qquad \mathbf{x} = (u, v, 1)^\top.$$

The 3×4 projection matrix is

$$P = K[R \mid \mathbf{t}].$$

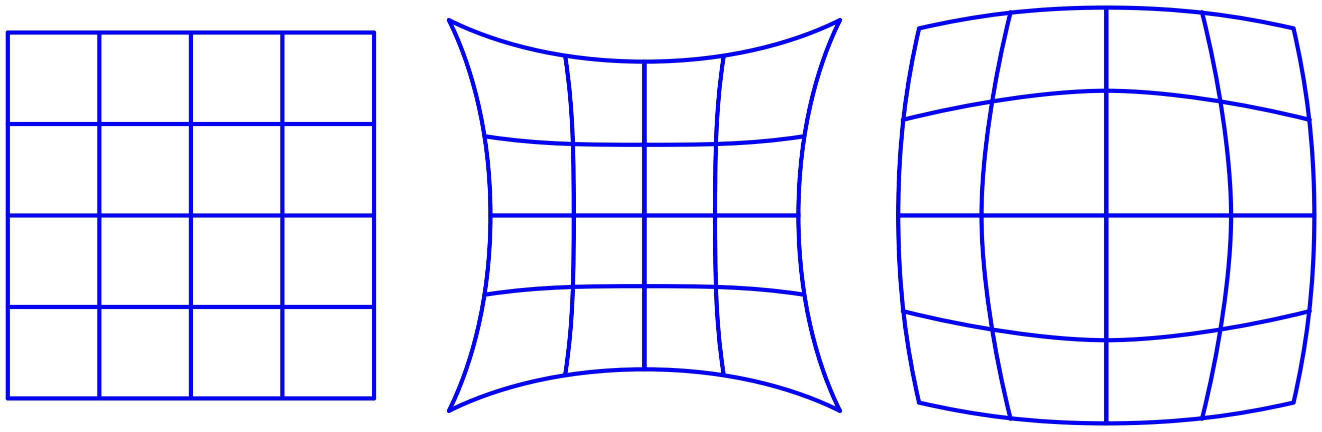
Projection Matrix Structure

$$P = \begin{pmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{pmatrix} = \begin{pmatrix} \mathbf{p}_1^\top \\ \mathbf{p}_2^\top \\ \mathbf{p}_3^\top \end{pmatrix}.$$

The image coordinates are

$$u = \frac{\mathbf{p}_1^\top \mathbf{X}}{\mathbf{p}_3^\top \mathbf{X}}, \qquad v = \frac{\mathbf{p}_2^\top \mathbf{X}}{\mathbf{p}_3^\top \mathbf{X}}.$$

Distortion: Why the Pinhole Model Is Not Enough



- Real lenses bend rays away from ideal central projection.
- Calibration usually estimates an ideal pinhole camera plus a distortion map.
- Distortion is strongest far from the principal point.

Radial Distortion

Let (x, y) be normalized image coordinates relative to the distortion center and $r^2 = x^2 + y^2$.

$$x_d = x\left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6 + \cdots\right),$$
$$y_d = y\left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6 + \cdots\right).$$

- Barrel distortion typically corresponds to negative first-order radial coefficient under this convention.
- Pincushion distortion typically corresponds to positive first-order radial coefficient.

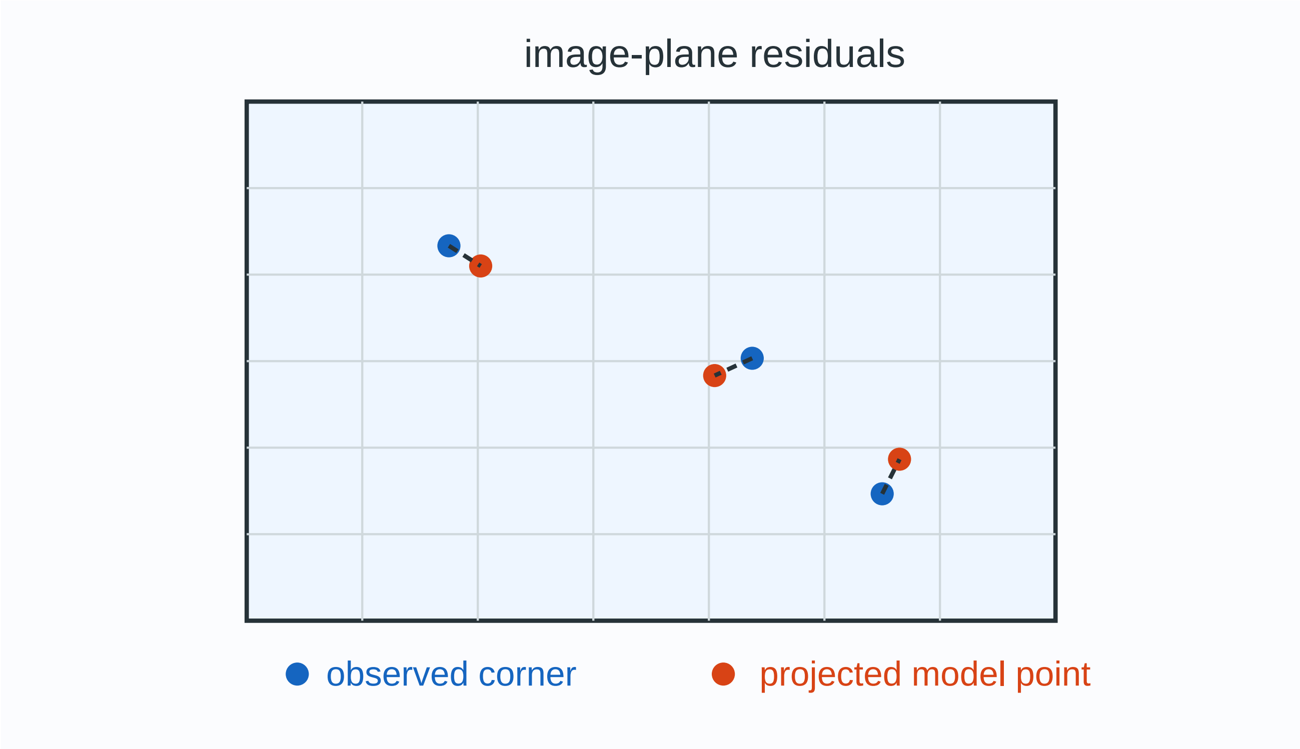
Tangential Distortion

Tangential distortion comes from lens decentering or imperfect alignment:

$$
\begin{aligned}
x_d &= x + 2p_1 xy + p_2(r^2 + 2x^2), \\
y_d &= y + p_1(r^2 + 2y^2) + 2p_2 xy.
\end{aligned}
$$

These terms are the standard Brown-Conrady/OpenCV form for normalized coordinates.

Reprojection Error



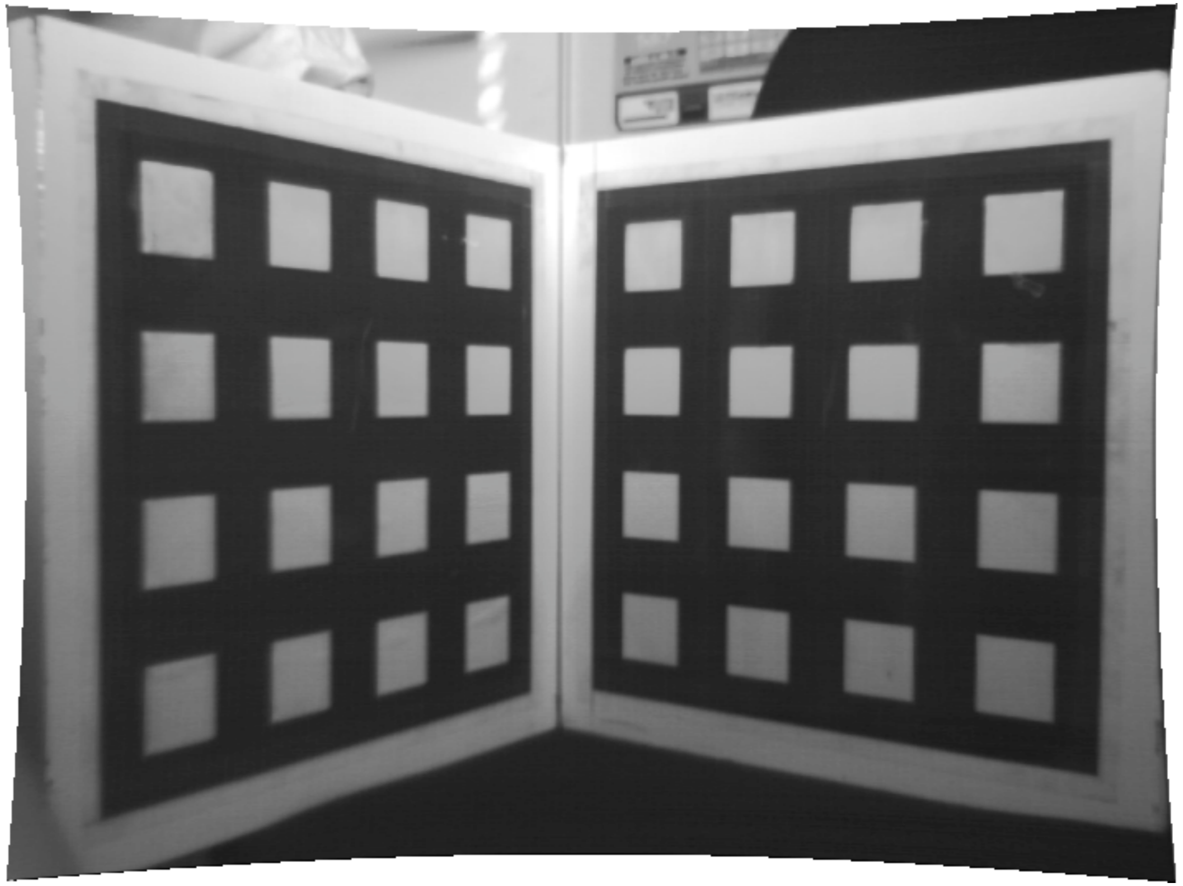
The geometric objective is image-plane residual:

$$\min_{\theta} \sum_{i=1}^{N} \|\mathbf{x}_i - \pi(P_\theta \mathbf{X}_i)\|_2^2.$$

Here $\pi(a, b, c)^\top = (a/c, b/c)^\top$.

Calibration

What Calibration Estimates

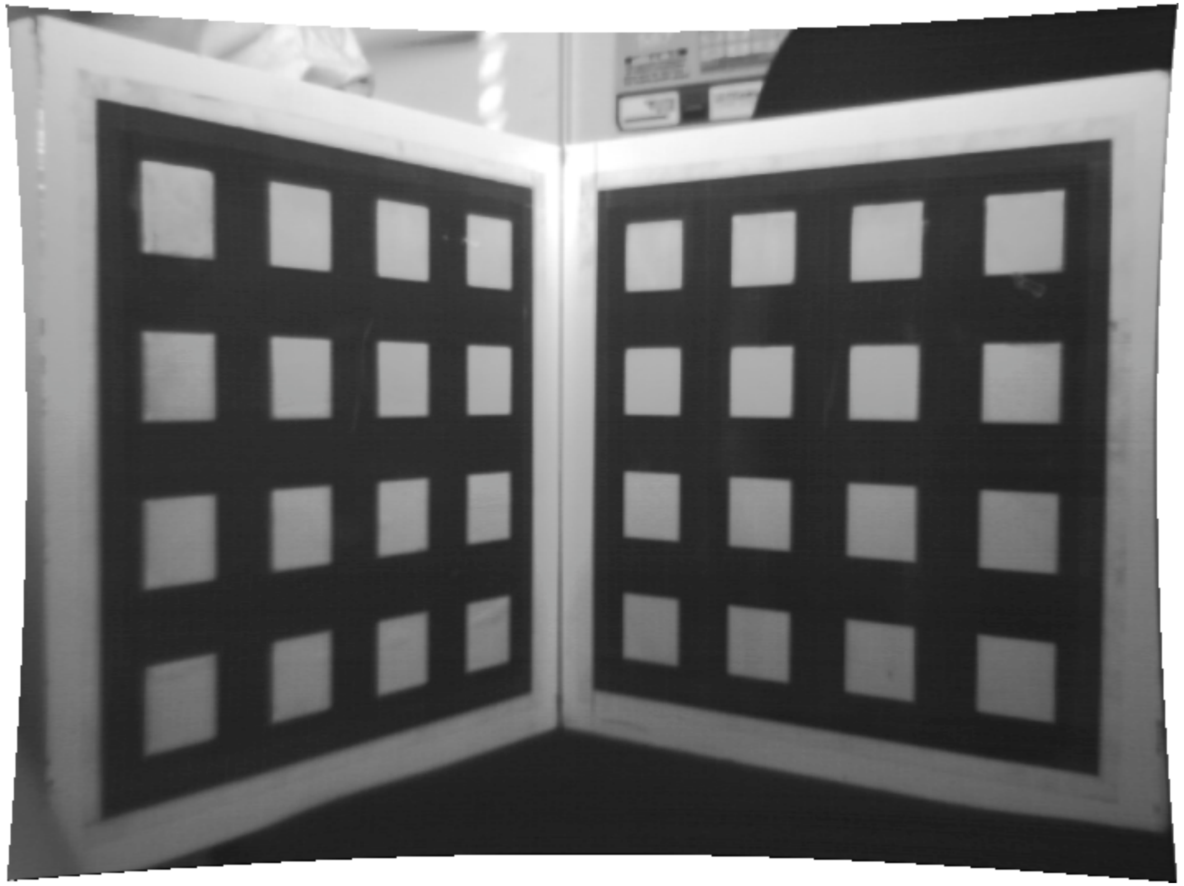


Tsai calibration is commonly introduced with a calibrated dot target whose 3D control-point coordinates are known.

Given target points $\mathbf{X}_i$ and detected image points $\mathbf{x}_i$, calibration estimates:

- intrinsics K
- camera pose $(R, \mathbf{t})$
- lens distortion parameters
- uncertainty or reprojection residuals

Tsai Calibration: Core Idea



- Staged pinhole-plus-distortion model.
- Linear constraints estimate first pose and scale.
- Additional equations recover focal length and depth translation.
- Nonlinear refinement minimizes reprojection error.

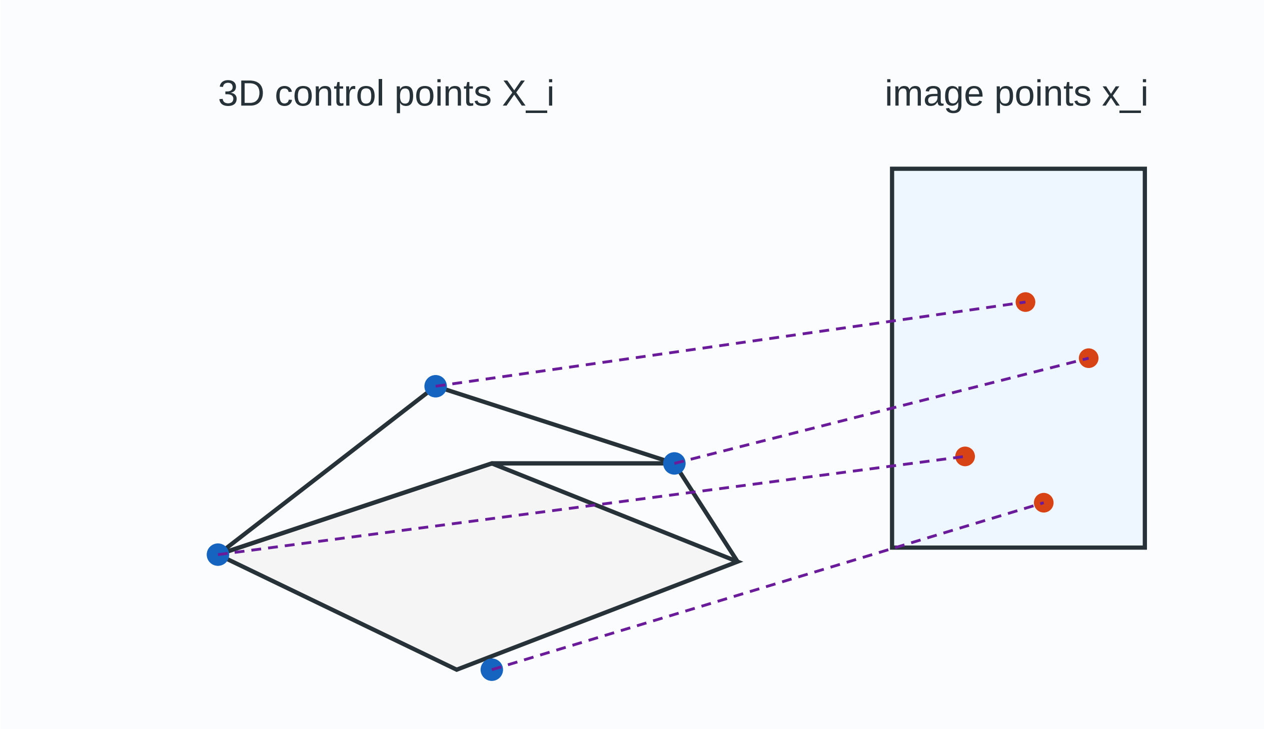
The staged estimate is an initialization; the final objective is geometric.

Calibration Pipeline

1. Detect calibration target corners.
2. Associate corners with known target coordinates.
3. Estimate an initial camera matrix or staged Tsai parameters.
4. Estimate lens distortion from residual structure.
5. Refine all parameters by nonlinear least squares.
6. Inspect reprojection error and residual patterns.

DLT

DLT Problem



Given N correspondences

$$\mathbf{X}_i = (X_i, Y_i, Z_i, 1)^\top, \qquad \mathbf{x}_i = (u_i, v_i, 1)^\top,$$

estimate the uncalibrated projection matrix P.

At least 6 nondegenerate correspondences are needed.

DLT Starts From Projective Equality

$$\mathbf{x}_i \sim P\mathbf{X}_i.$$

Equivalently,

$$u_i = \frac{\mathbf{p}_1^\top \mathbf{X}_i}{\mathbf{p}_3^\top \mathbf{X}_i},$$
$$v_i = \frac{\mathbf{p}_2^\top \mathbf{X}_i}{\mathbf{p}_3^\top \mathbf{X}_i}.$$

Cross-multiplication removes the division.

DLT Linear Equations

Each correspondence gives two independent linear equations:

$$\begin{aligned} \mathbf{p}_1^\top \mathbf{X}_i - u_i \mathbf{p}_3^\top \mathbf{X}_i &= 0, \\ \mathbf{p}_2^\top \mathbf{X}_i - v_i \mathbf{p}_3^\top \mathbf{X}_i &= 0. \end{aligned}$$

These are linear in the 12 entries of P.

DLT Matrix Rows

Let $\mathbf{p}$ be the 12-vector formed by stacking the rows of P. The two rows for correspondence i are:

$$\mathbf{a}_{2i-1} = \begin{bmatrix} \mathbf{X}_i^\top & \mathbf{0}^\top & -u_i \mathbf{X}_i^\top \end{bmatrix},$$
$$\mathbf{a}_{2i} = \begin{bmatrix} \mathbf{0}^\top & \mathbf{X}_i^\top & -v_i \mathbf{X}_i^\top \end{bmatrix}.$$

Stack the System

Stacking all rows gives

$$A\mathbf{p} = \mathbf{0}, \qquad A \in \mathbb{R}^{2N\times 12}.$$

Equivalently,

$$A = \begin{pmatrix} \mathbf{X}_1^\top & \mathbf{0}^\top & -u_1\mathbf{X}_1^\top \\ \mathbf{0}^\top & \mathbf{X}_1^\top & -v_1\mathbf{X}_1^\top \\ \vdots & \vdots & \vdots \\ \mathbf{X}_N^\top & \mathbf{0}^\top & -u_N\mathbf{X}_N^\top \\ \mathbf{0}^\top & \mathbf{X}_N^\top & -v_N\mathbf{X}_N^\top \end{pmatrix}.$$

Because P is defined up to scale, solve for a nonzero null vector.

SVD Solution

Solve the constrained least-squares problem:

$$\mathbf{p}^\star = \arg\min_{\|\mathbf{p}\|_2=1} \|A\mathbf{p}\|_2.$$

If $A = U\Sigma V^\top$, then $\mathbf{p}^\star$ is the last column of V.

Reshape $\mathbf{p}^\star$ into $P \in \mathbb{R}^{3\times 4}$.

DLT Conditioning

- Raw pixel and world coordinates can have very different scales.
- Normalize image and world points before solving:
  - translate centroids to the origin
  - scale average distance to a fixed value
- Denormalize the recovered projection matrix afterward.
- This is the normalized DLT idea emphasized in Multiple View Geometry.

DLT Degeneracies

DLT becomes singular or unstable when correspondences do not constrain a full 3D projective camera.

Common failure modes:

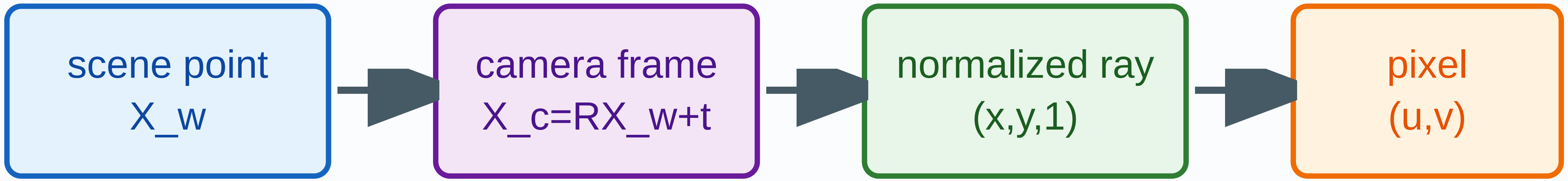
- all 3D points lie on a critical configuration
- all points are coplanar for a full 3×4 camera estimate
- too few points or nearly duplicate points
- gross outliers in detected image locations

DLT in Practice

- DLT is a direct initializer, not usually the final calibration.
- Refine P or $(K, R, \mathbf{t})$ with reprojection-error minimization.
- If calibration structure is known, enforce it after the linear step.
- Use robust estimation when correspondences may contain outliers.

Summary

Complete Camera Pipeline



$$x \sim K\,[R \mid t]\,X$$

calibration estimates K, pose estimates R,t, reconstruction inverts rays with multi-view constraints

Takeaways

- Perspective projection is linear in homogeneous coordinates and nonlinear after dehomogenization.
- $P = K[R \mid \mathbf{t}]$ combines intrinsics, pose, and perspective division.
- The camera center satisfies $P(\mathbf{C}, 1)^\top = \mathbf{0}$.
- Calibration estimates the model by minimizing image-plane reprojection error.
- DLT builds a homogeneous linear system from 3D–2D correspondences.

References

- Hartley and Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed.
- MIT VisionBook, Chapter 39: Camera Modeling and Calibration.
- Cyrill Stachniss, Camera Calibration / DLT lecture materials.
- Wikipedia/Wikimedia Commons pages for pinhole camera model, camera matrix, perspective projection, DLT, and lens distortion diagrams.